

Project Executable Files

Date	01-07-2026
Team ID	SWTID-2026-3845
Project Name	SMART LENDER
Maximum Marks	3 Marks

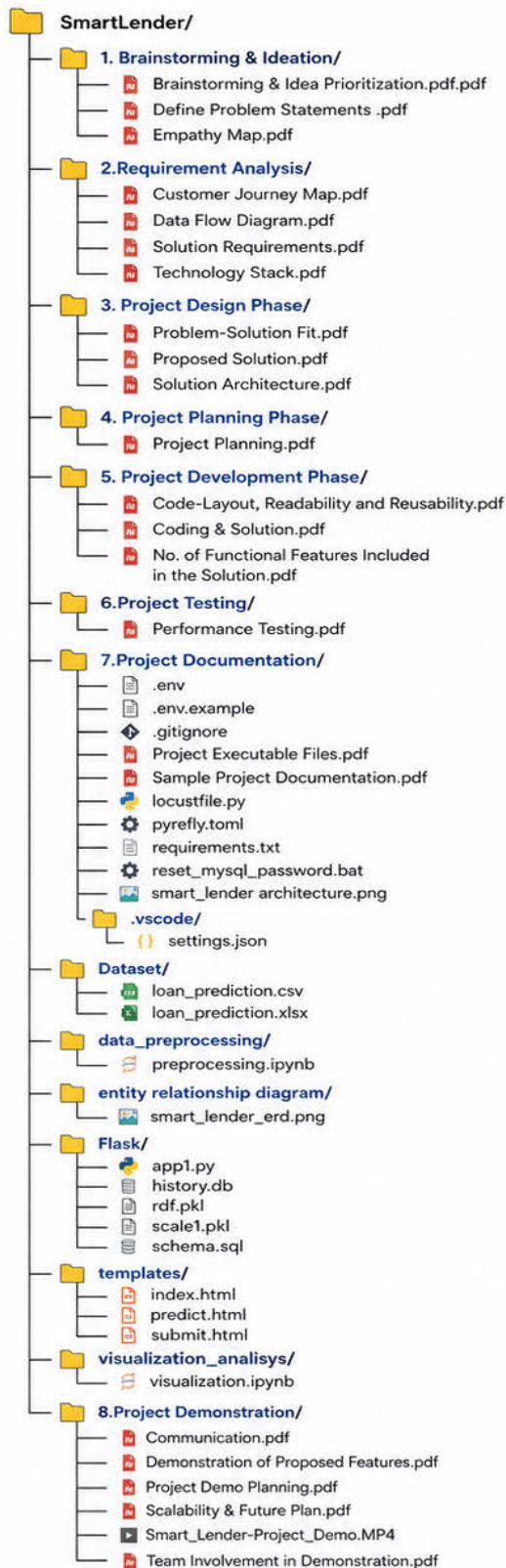
Project Executable Files

This section requires submission of all executable and deployable files for your project. Ensure all files are properly organized, named, and uploaded to the specified submission platform. The evaluator must be able to run or deploy your solution using the provided files and instructions.

Step 1: Submission Checklist

S.No	Item to Submit	Submitted(Yes/No / NA)
1	Complete source code (all files and folders)	Yes
2	README / Setup Guide (instructions to run the project)	Yes
3	requirements.txt / package.json / dependency file	Yes
4	Database schema / seed files (if applicable)	Yes
5	Environment configuration file (.env.example or similar)	Yes
6	Deployed application URL (if hosted)	Yes
7	APK / Executable binary (if applicable for mobile/desktop apps)	No
8	Dockerfile / Containerization config (if applicable)	No
9	Test files and test results	Yes
10	Demo video or walkthrough (if required)	Yes

SmartLender – Project Folder Structure



Folder / File	Description
1. Brainstorming & Ideation/	Documents for brainstorming and idea generation
2. Requirement Analysis/	Requirement gathering and analysis documents
3. Project Design Phase/	Design and architecture planning documents
4. Project Planning Phase/	Project planning documents
5. Project Development Phase/	Development phase documents
6. Project Testing/	Testing reports and documents
7. Project Documentation/	Project documentation and configuration files
.env	Environment variables file
.env.example	Example environment file
.gitignore	Git ignore configuration
Project Executable Files.pdf	Executable files guide
Sample Project Documentation.pdf	Sample documentation
locustfile.py	Load testing configuration file
pyrefly.toml	Python project configuration
requirements.txt	Python dependencies file
reset_mysql_password.bat	Script to reset MySQL password
smart_lender architecture.png	High-level architecture diagram
.vscode/	VS Code workspace settings
Dataset/	Raw loan prediction datasets (CSV & Excel)
data_preprocessing/	Data preprocessing notebooks
entity relationship diagram/	Database ER diagram
Flask/	Flask application and database files
templates/	HTML templates for the web app
visualization_analysys/	Data analysis and visualization notebooks
8. Project Demonstration/	Demo planning and presentation materials

Step 3: Deployment / Access Details

Field	Details
Hosted / Deployed URL	https://smart-lender-dple.onrender.com
Login Credentials (Demo)	None
Platform / Hosting Provider	Render
Repository Link	https://github.com/kaladharroyal/smart-lender
Demo Video Link	https://drive.google.com/file/d/1k_6UA40vzo-iLIYRRP-XgcY9YUhaNkyW/view?usp=drivesdk

Pre-requisites:

- Python 3.8 or higher
- pip (Python package manager)
- Git
- Web browser (Chrome/Firefox)
- Required libraries: Flask, Pandas, NumPy, Scikit-learn, XGBoost, Pickle/Joblib

Step 4: Run Instructions

Provide clear, step-by-step instructions for the evaluator to run your project locally:

Pre-requisites:

Python 3.8 - 3.12, MySQL Server (v8.0 or newer), pip (Python package manager).

Step 1: Create a virtual environment: `python -m venv .venv`

Step 2: Activate the virtual environment (Windows PowerShell): `.venv\Scripts\Activate.ps1`

Step 3: Install dependencies: `pip install -r requirements.txt`

Step 4: Log in to MySQL and create the database: `CREATE DATABASE IF NOT EXISTS smart_lender;`

Step 5: Import the database schema: `mysql -u root -p smart_lender < Flask/schema.sql`

Step 6: Create a `.env` file in the root directory (use `.env.example` as reference):

```
DB_HOST=127.0.0.1
DB_PORT=3306
DB_USER=root
DB_PASSWORD=your_mysql_password
DB_NAME=smart_lender
```

Step 7: Run the Flask application: `python Flask/app1.py`

Step 8: Open browser at `http://127.0.0.1:5000/`

Step 5: Known Issues / Limitations

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S.No	Known Issue / Limitation	Workaround / Status
1	Model test accuracy is capped at ~84.9% (XGBoost); some misclassifications occur for borderline applicants.	Acceptable for demo/prototype use; improvement would need more data and feature engineering.
2	Dataset is small (614 records) and balanced using SMOTE, which generates synthetic samples rather than real data.	May not generalize to larger real-world populations; retraining on live data recommended before production.
3	App is hosted on Render's free tier, which spins down when idle, causing a cold-start delay on first request.	Acceptable for demo; a paid/always-on tier would resolve this in production.
4	No user authentication/login system; anyone with the URL can submit applications or view the dashboard.	Suitable for demo/evaluation only; role-based access control needed before real deployment.
5	MySQL database is a single instance with no replication or backup strategy.	Fine for prototype; production use needs managed DB hosting with backups.
6	Outlier handling (IQR) and missing-value imputation (mode) are simple statistical methods.	Works for current dataset; more advanced techniques could be explored in future iterations.